

Rémi Imbach

Curriculum Vitæ

Civil status: Rémi Imbach, born December 7, 1985 in Strasbourg
French citizen, married, one child.

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Abstract

Last positions

- 2023 - now **Research Engineer**, IRMA¹, Université de Strasbourg (UdS), then IN-RIA² Nancy - Grand Est, France
- 2022 - 2023 **Research Engineer** INRIA Nancy - Grand Est, France
- 2020 - 2022 **Consultant**, independent; contracts with Waterloo Maple Incorporation (WMI), Canada, and European Science Foundation, France
- 2018 - 2020 **Assistant Professor**, Courant Institute of Mathematical Sciences (CIMS), New York University (NYU), New York, USA

Education

2013 **PhD in Computer Science**, UdS, France

Recent software

- 2025 **ScimBa**: Python library for Physics Informed Machine Learning. Backends Pytorch or Jax.
- 2023 **pwpoly**: C library for fast polynomial evaluation and root-finding, Python wrapper and Maple 2024³ module

Keywords

Scientific: Physics Informed Machine Learning, numeric and symbolic computation, certified computation, polynomial root finding, computer arithmetic

Technologies: Jax, Pytorch, Python, C, C++, CMake, Maple, git, L^AT_EX, CI, Linux

¹Research Institute for Advanced Mathematics

²National Institute for Research in Digital Science and Technology

³<https://www.maplesoft.com/>

Positions

since dec. 2024	Research Engineer , INRIA Nancy - Grand Est, France <i>Developping ScimBa library</i> , MACARON ⁴ team software project: Scimba
2024 (10 mois)	Research Engineer , IRMA, UdS, France, <i>Radiative transfer, direct and inverse problem for water turbidity measurment</i> , MOCO ⁵ team software projec: AnaTrot (not published)
2022 - 2023 18 months	Research Engineer , INRIA Nancy - Grand Est, France <i>Implementation of a root-finding algorithm, integration in Maple</i> part of GAMBLE ⁶ team, work with G. Moroz (INRIA) Publication: [1] Software: HEFRoots, befft, PWPoly
2020 - 2022 20 months	Consultant , independent; contracts with V. Pan (CUNY ⁷), with WMI and with European Science Foundation ⁸ . Publications: [2, 3] Software: Ccluster, HEFRoots
2018 - 2020 28 months	Assistant Professor (without tenure), CIMS, NYU, USA <i>Research in symbolic computation and root-finding</i> with C. Yap and V. Pan Publications: [4, 5, 6, 7] Software: Ccluster Teaching: 125 hours
2017-2018 10 months	Scientific assistant , Technische Universität Kaiserslautern (TUK), Germany <i>Implementation of a near optimal root clustering algorithm</i> Part of AGAG team Publication: [8] Software: Ccluster
2014 - 2016 24 months	Postdoctoral Researcher , INRIA Nancy - Grand Est, France <i>Certified numerical algorithms for computing topology of curves and surfaces</i> Advisors: G. Moroz and M. Pouget Publications: [9, 10, 11, 12, 13] Software: subdivision_solver
2013 - 2014 11 months	Research and teaching fellow , UdS, ICube ⁹ laboratory, France <i>Reparameterization and homotopy for point distance constraint problems</i> Advisors: P. Mathis and P. Schreck, UdS Publication: [14] Teaching: 180 hours
2010 - 2013 36 months	PhD candidate , UdS, ICube lab., France <i>Solving geometric constraints by leading an homotopy method by geometry</i> Advisor: P. Mathis; Supervisor: P. Schreck Publications: [15, 16, 17, 18] PhD thesis: [19] Teaching: 3x64 hours

⁴MACHine leARning for Optimized Numerical methods

Software development

2024 -	ScimBa ¹⁰ : Python library, for Scientific Machine Learning. It offers cutting edge approaches for approximating solutions of PDE based problems such as PINNs, DeepRitz, DeepFlows, NeuralSemiLagrangian using efficient classical (SS-BFGS, SS-Broyden) and physics informed optimizer (natural gradients). Two backends: Jax and Pytorch.
2024	AnaTrot Python code based on Pytorch for radiative transfer direct and inverse problem.
2022 - 2023	PWPoly ¹¹ : C library implementing algorithms for root finding and fast evaluation of univariate polynomials ([1]), with Python interface available in Maple 2024, <code>fsolve</code> ¹² command or <code>RootFinding:-Isolate</code> ¹³ , HR method $\simeq 20\,000$ lines of code
2022 - 2023	befft ¹⁴ : C library with Python interface for radix 2 FFT in machine doubles with guaranteed error bounds
2021 - 2022	HEFRoots : (first version of PWPoly) C library implementing a clustering algorithm ¹⁵ . Available in Maple 2023, <code>fsolve</code> ¹⁶ command or <code>RootFinding:-Isolate</code> ¹⁷ , HR method $\simeq 30\,000$ lines of code
2018 - 2020	Ccluster ¹⁸ : C library implementing a near optimal algorithm for root clustering ¹⁹ . Wrapper for Julia, (<code>Ccluster.jl</code> ²⁰). Cited in [8, 3, 5]. $\simeq 35\,000$ lines of code
2014 - 2016	subdivision_solver ²¹ : SageMath package for polynomial system solving. Designed for systems of huge polynomials (in <i>bitsize</i> and <i>degree</i>). Written in C++, Cython. Described in [12]. $\simeq 8\,000$ lines of code

⁵Model and Control

⁶Geometric Algorithms & Models Beyond the Linear & Euclidean realm

⁷City University of New York

⁸<https://www.esf.org>

⁹Engineering science, computer science and imaging laboratory

¹⁰<https://www.scimba.org>

¹¹<https://gitlab.inria.fr/gamble/pwpoly>

¹²<https://www.maplesoft.com/support/help/maple/view.aspx?path=fsolve%2fdetails>

¹³[https://www.maplesoft.com/support/help/maple/view.aspx?path=RootFinding%](https://www.maplesoft.com/support/help/maple/view.aspx?path=RootFinding%2fIsolate)

[2fIsolate](https://www.maplesoft.com/support/help/maple/view.aspx?path=RootFinding%2fIsolate)

¹⁴<https://gitlab.inria.fr/gamble/befft>

¹⁵Guillaume Moroz. *New data structure for univariate polynomial approximation and applications to root isolation, numerical multipoint evaluation, and other problems*. 2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)

¹⁶<https://www.maplesoft.com/support/help/maple/view.aspx?path=fsolve%2fdetails>

¹⁷[https://www.maplesoft.com/support/help/maple/view.aspx?path=RootFinding%](https://www.maplesoft.com/support/help/maple/view.aspx?path=RootFinding%2fIsolate)

[2fIsolate](https://www.maplesoft.com/support/help/maple/view.aspx?path=RootFinding%2fIsolate)

¹⁸<https://github.com/rimbach/Ccluster>

¹⁹Ruben Becker, Michael Sagraloff, Vikram Sharma, Juan Xu and Chee Yap. *Complexity Analysis of Root Clustering for a Complex Polynomial*. Proceedings of the ACM on International Symposium on Symbolic and Algebraic Computation, ISSAC '16

²⁰<https://github.com/rimbach/Ccluster.jl>

Recent communications

- June 2026 *Workshop on Machine Learning and Automatic Differentiation in JAX for Scientific Computing*, Strasbourg²².
tutorial https://rimbach.gitlabpages.inria.fr/ml_ad_jax_sc_pinns_practical/
- May 2026 Summer School *Machine Learning + Simulation*, Sophia Antipolis²³.
- Dec. 2025 Workshop *Fabrique Ton PINNs*²⁴, Sorbonne Université, Paris. Organized by Groupe Calcul, CNRS.
tutorial <https://rimbach.gitlabpages.inria.fr/fabrique-ton-pinns/>

Teaching experiences

- 2024 **Teaching fellow**, UFR MAI²⁵, Licence, 35h contact
- Summer 2018 to Spring 2020 **Responsible and main instructor** of *Mathematical Techniques for Computer Science Applications*, Master's students in Computer Science, CIMS, NYU, 125 hours lectures.
Syllabus: Linear algebra, probability, numeric methods for signal processing, computer graphics and machine learning. 1 grader, 1 teaching assistant.
- 2013 - 2014 **Teaching fellow**, UdS: 170 hours exercises sessions, 10 hours lectures
Responsible of *Algorithm and Coding*, under-graduate students in mathematics, physics and chemistry; 10h lectures in this course unit.
Exercises sessions in *graph theory, computers architecture, ...*
- 2010 - 2014 **Adjunct professor**, UdS: 3×64 hours exercises sessions in *theory of operating systems, computers architecture, graph theory, algorithms and coding, ...*

Education

- October 2013 **PhD** in Computer Science, UdS, France
Solving geometric constraints by leading an homotopy method by geometry
Advisor: P. Mathis; Supervisor: P. Schreck
Committee members: D. Michelucci, B. Mourrain, M.Tajine, P. Serré.
- June 2010 **Master's degree** in Computer Science, UdS
Specialty: Computer Science and Imaging
- September 2008 **Master's degree** in Mathematics, UdS
Specialty: Discrete Mathematics

²¹<https://gitlab.inria.fr/gamble/subdiv-solver>

²²<https://majsc2026.pages.math.unistra.fr/index.html>

²³<https://project.inria.fr/mlsim2026/summer-school/>

²⁴<https://groupe-calcul.cnrs.fr/2025-12-18-atelier-pinns.html>

²⁵Mathématiques et d'Informatique

Referees

Dr. Emmanuel Franck (`emmanuel.franck@inria.fr`) is researcher at INRIA Nancy-Grand Est, France

Dr. Guillaume Moroz (`guillaume.moroz@inria.fr`) is researcher at INRIA Nancy-Grand Est, France
INRIA Nancy-Grand Est 615 rue du jardin botanique 54600 Villers lès Nancy
+33 (0)3 54 95 84 79

Dr. Jürgen Gerhard (`jgerhard@maplesoft.com`) is Senior director of research at Maplesoft, WMI

Pr. Victor Y. Pan (`victor.pan@lehman.cuny.edu`) is Distinguished professor of Math. at CUNY.
Graduate Center of the CUNY, 365 Fifth Avenue, New York, NY 10016-4309, USA

Scientific diffusion

Publications

In the community of symbolic computing where [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 20] have been published, the usage is to make appear the names of the authors in their alphabetical order.

In the community of computer graphics where [14, 15, 16, 17, 18] have been disseminated, the usage is to make appear the name of the main contributor in first position.

International journals

- [1] Rémi Imbach, and Guillaume Moroz. Fast evaluation and root finding for polynomials with floating-point coefficients. *Journal of Symbolic Computation*, 127, 2025
- [5] Rémi Imbach, Marc Pouget and Chee Yap. Clustering Complex Zeros of Triangular Systems of Polynomials. *Mathematics in Computer Science*, 1–22, 2020 <https://doi.org/10.1007/s11786-020-00482-0>
- [9] Rémi Imbach, Guillaume Moroz, and Marc Pouget. Reliable location with respect to the projection of a smooth space curve. *Reliable Computing*, 26:13 – 55, 2018. https://interval.louisiana.edu/reliable-computing-journal/tables-of-contents.html#Volume_26
- [10] Rémi Imbach, Guillaume Moroz, and Marc Pouget. A certified numerical algorithm for the topology of resultant and discriminant curves. *Journal of Symbolic Computation*, 80, Part 2:285 – 306, 2017 <http://dx.doi.org/10.1016/j.jsc.2016.03.011>
- [14] Rémi Imbach, Pascal Mathis, and Pascal Schreck. A robust and efficient method for solving point distance problems by homotopy. *Mathematical Programming*, 163:115–144, 2017 <https://doi.org/10.1007/s10107-016-1058-7>
- [15] Rémi Imbach, Pascal Schreck, and Pascal Mathis. Leading a continuation method by geometry for solving geometric constraints. *Computer-Aided Design*, 46:138–147, 2014 <https://doi.org/10.1016/j.cad.2013.08.026>

International conferences, peer reviewed

- [2] Rémi Imbach and Victor Y. Pan. Accelerated Subdivision for Clustering Roots of Polynomials Given by Evaluation Oracles. In *Computer Algebra in Scientific Computing: 24th International Workshop, CASC 2022*, 143–164, 2022, Springer. https://link.springer.com/chapter/10.1007/978-3-031-14788-3_9
- [3] Rémi Imbach and Victor Y. Pan. Root Radii and Subdivision for Polynomial Root-Finding. In *Computer Algebra in Scientific Computing: 23rd International Workshop, CASC 2021*, 136–156, 2021, Springer. https://link.springer.com/chapter/10.1007/978-3-030-85165-1_9
- [4] Rémi Imbach and Victor Y. Pan. New progress in univariate polynomial root finding. In *International Symposium on Symbolic and Algebraic Computation, ISSAC 2020*, page 249–256, 2020, Association for Computing Machinery. <https://doi.org/10.1145/3373207.3404063>

- [6] Rémi Imbach and Victor Y. Pan. New practical advances in polynomial root clustering. In *Mathematical Aspects of Computer and Information Sciences, MACIS 2019*, pages 122–137, Cham, 2019. Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-030-43120-4_11
- [7] Rémi Imbach, Victor Y. Pan, Chee Yap, Ilias S. Kotsireas, and Vitaly Zaderman. Root-finding with implicit deflation. In *Computer Algebra in Scientific Computing, CASC 2019*, pages 236–245, Cham, 2019. Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-030-26831-2_16
- [8] Rémi Imbach, Victor Y. Pan, and Chee Yap. Implementation of a near-optimal complex root clustering algorithm. In *Mathematical Software, ICMS 2018*, pages 235–244, Cham, 2018. Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-319-96418-8_28
- [11] Rémi Imbach, Guillaume Moroz, and Marc Pouget. Numeric and Certified Isolation of the Singularities of the Projection of a Smooth Space Curve. In *Mathematical Aspects of Computer and Information Sciences: 6th International Conference, MACIS 2015*, pages 78–92. Springer International Publishing, Cham, 2016. http://dx.doi.org/10.1007/978-3-319-32859-1_6
- [16] Pascal Mathis, Pascal Schreck, and Rémi Imbach. Decomposition of geometrical constraint systems with reparameterization. In *Proceedings of the 27th Annual ACM Symposium on Applied Computing, SAC 2012*, pages 102–108. ACM, 2012. <http://dl.acm.org/citation.cfm?id=2245298>
- [18] Rémi Imbach, Pascal Mathis, and Pascal Schreck. Tracking method for reparametrized geometrical constraint systems. In *13th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNASC 2011*, pages 31–38. IEEE, 2011. <https://ieeexplore.ieee.org/abstract/document/6169498>

Research report

- [20] Rémi Imbach and Victor Y. Pan. Polynomial root clustering and explicit deflation. arXiv preprint arXiv:1906.04920, June 2019. <https://arxiv.org/abs/1906.04920>

Technical report

- [12] Rémi Imbach. A Subdivision Solver for Systems of Large Dense Polynomials. Technical Report RT-0476, INRIA Nancy, March 2016. <https://arxiv.org/abs/1603.07916>

National conferences, non peer reviewed

- [13] Rémi Imbach, Guillaume Moroz, and Marc Pouget. A certified numerical approach to describe the topology of projected curves. In *Journées de l'Association Française d'Informatique Graphique*, 2015.
- [17] Rémi Imbach, Pascal Mathis, and Pascal Schreck. Une approche par décomposition et reparamétrisation de systèmes de contraintes géométriques. In *Journées du Groupe de Travail en Modélisation Géométrique*, 2012.

PhD Thesis

- [19] Rémi Imbach. *Résolution de contraintes géométriques en guidant une méthode homotopique par la géométrie*. PhD thesis, Université de Strasbourg, 2013. <https://tel.archives-ouvertes.fr/tel-01124176/>

Selected communications

Seminars

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|------------|---|
| Jan. 2021 | <i>Complex Roots Clustering</i> , Séminar PolSys-SpecFun ²⁶ , Paris |
| Nov. 2019 | <i>Practical Advances in Complex Root Clustering</i> .
Joint CUNY Graduate Center-Courant Seminar in Symbolic-Numeric Computing, Courant Institute of Mathematical Sciences, New York. ²⁷ |
| Jan. 2019 | <i>Complex Roots/Solutions Clustering Algorithms</i> .
OGRE team seminar, Nantes, France |
| May 2018 | <i>Numerical and certified computation of the topology of projected curves</i> .
Joint CUNY Graduate Center-Courant Seminar in Symbolic-Numeric Computing, CUNY Graduate Center, New York. |
| July 2017 | <i>Certified numerical tools for computing the topology of projected curves</i> .
AGAG seminar, Technische Universität Kaiserslautern Germany ²⁸ |
| Sept. 2016 | <i>Certified numerical tools for computing the topology of projected curves</i> .
AriC seminar, ENS(École Normale Supérieure) Lyon, France ²⁹ |

²⁶<https://polysys.lip6.fr/~safey/seminar.html>

²⁷<http://qcpages.qc.cuny.edu/~aovchinnikov/seminar.html>

²⁸<http://www.mathematik.uni-kl.de/en/agag/talks/20170101/>

²⁹<http://www.ens-lyon.fr/LIP/AriC/seminar>

International conferences and workshops

July 2023	<i>Fast evaluation and root finding for polynomials with floating-point coefficients.</i> ISSAC 2023 (International Symposium on Symbolic and Algebraic Computation), Tromsø, Norway conference ³⁰ .
July 2020	<i>New progress in univariate polynomial root finding.</i> ISSAC 2020, online conference ³¹ .
Aug. 2019	<i>Clustering Complex Zeros of Triangular Systems of Polynomials.</i> CASC 2019 (Computer Algebra in Scientific Computing), Moscow, Russia ³² .
July 2018	<i>Implementation of a near-optimal complex root clustering algorithm.</i> ICMS 2018 (International Congress of Mathematical Software), Notre-Dame, USA ³³
June 2016	<i>Interval tools for computing the topology of projected curves.</i> SWIM 2016 (Summer Workshop on Interval Methods), Lyon, France ³⁴
Novembre 2015	<i>Numeric and Certified Isolation of the Singularities of the Projection of a Smooth Space Curve.</i> MACIS 2015 (Sixth International Conference on Mathematical Aspects of Computer and Information Science), Berlin, Germany ³⁵
Novembre 2013	<i>Leading a continuation method by geometry for solving geometric constraints.</i> GD/SPM 13 (Geometric and Physical Modeling), Denver, Colorado, USA
Septembre 2011	<i>Tracking method for reparametrized geometrical constraint systems.</i> SYNASC 11 (Symposium on Symbolic and Numeric Algorithms for Scientific Computing), Timisoara, Romania

³⁰<http://www.issac-conference.org/2023/>

³¹<http://www.issac-conference.org/2020/>

³²<http://www.casc-conference.org/2019/>

³³<http://icms-conference.org/2018/>

³⁴<https://swim2016.sciencesconf.org/>

³⁵<http://macis2015.zib.de/>